API и интерфейс

API → состояние страницы → сравнение → сценарии → демо

Что нужно знать

- Почему FastAPI читает подготовленные таблицы вместо выдачи миллионов исходных строк в браузер.
- Состояние league/season/teams в URL; проверка параметров, read-only pool и ограничение времени SQL-запроса.
- Как считается очное сравнение с позиции каждого клуба: дома/в гостях меняются goals_for и goals_against.
- При отсутствии матчевого CSV показывается отсутствие покрытия; это не доказательство отсутствия встреч и не 0–0.
- Как Lab получает точные пять входов для клуба/сезона/соревнования, ограничивает attacking_spend общим бюджетом и отбрасывает устаревшие ответы.
- Что проверяют unittest, live API checks и реальный браузер; как читать /api/health и почему наличие одной таблицы недостаточно.
- Serie A 24/25 comparison: Inter 79 GF, 35 GA, +44 GD, attack rank 1 and defense rank 2; Milan 61, 43, +18, attack rank 3 and defense rank 7. Ranks are by per-match scored/conceded goals in the league-season.
- xG is absent from the actual loaded match schema. Do not infer it from goals or player aggregates.
- The optional player source has goalkeeper conceded totals. They exactly match the actual scoreboard GA in 366 of 388 complete top-five club-seasons; clean-sheet totals agree in 269. The UI uses actual scorelines for league ranks.

Что показать

1. Выбрать Serie A 24/25, открыть Inter/Milan и показать 1–2, 1–1, суммарные голы 2–3.
2. Показать, что без обогащения интерфейс даёт понятное сообщение, а не выдуманные значения.
3. Открыть Lab, изменить один вход, дождаться ответа и выполнить Reset. Не тратить время демо на все слайдеры.

Короткая речь на английском

> *The browser reads a bounded API over prepared results. League and season filters are explicit, and the URL preserves the selected scope. In the comparison, season totals and actual head-to-head meetings are separate. Serie A 24/25 shows Inter at 79 for and 35 against (+44), ranked first in goals scored per match and second in goals conceded per match. Milan has 61 for and 43 against (+18), ranked third and seventh. The source has no xG field. For Inter and Milan in 24/25, the recorded league scores are 1–2 and 1–1, giving head-to-head goals of two to three. Missing match enrichment is reported explicitly. The Lab loads the five real club-season inputs, calls a saved model and rejects invalid budgets or non-finite numbers. We verify behavior at three levels: regression tests, live data contracts and desktop/mobile browser checks.*

Переход: “The final question is whether this system runs reproducibly and what resources it uses.”

Практические задания

1. Поменяй клубы местами в URL и объясни, какие метрики должны поменять столбцы, а какие остаться равными.
2. Объясни различия HTTP 422, 404 и 503 на примерах этого API.
3. Проведи трёхминутное демо без чтения README: обзор → сравнение → Lab → Project.

Вопросы по направлению

Why have an API instead of plotting raw tables directly?

Зачем API?

The API exposes small prepared outputs under an explicit league and season. This avoids moving millions of raw rows to the browser, keeps SQL and validation on the server and separates presentation from processing.

How are head-to-head goals calculated?

Как считаются голы друг другу?

For each completed meeting, the selected club’s perspective determines whether the home or away score is goals for. Reversing the compared clubs swaps their totals. Wins, draws, losses and points are derived from those actual scores.

Does no recorded match mean a 0–0 draw?

Нет данных — это 0–0?

No. No coverage, no recorded meetings and a completed 0–0 draw are different states. The API reports enrichment availability and the interface shows an explicit message instead of inventing a score.

What prevents invalid scenario inputs?

Как защищены входы Lab?

The API rejects non-finite numbers, checks bounds and requires attacking spend not to exceed total spend. The browser also clamps the sliders. Server-side validation remains necessary even when the UI enforces the same rule.

What do HTTP 422, 404 and 503 mean here?

Что означают ошибки API?

422 indicates invalid input, such as selecting the same club twice. 404 means the requested club-season scope is unavailable. 503 indicates temporary data/model unavailability or an incomplete application output state.

What do your tests prove?

Что доказывают тесты?

Regression tests cover known failure cases, live checks validate the prepared outputs and API contracts, and browser checks exercise actual flows and mobile layout. They provide scoped evidence, not a guarantee that every possible bug is absent.

What is the main result in one sentence?

Главный результат одним предложением?

The system is reproducible on the measured setup, while historical transfer spending alone has almost no linear relationship with next-season index change and the richer model adds only modest predictive value.

What is the most important limitation?

Главное ограничение?

The data are retrospective and unevenly covered, the index is a proxy, and associations are not causal effects. Source licence and independent team/laptop validation also remain explicit unresolved requirements.

How do goals for, goals against and goal difference work in the comparison?

Как в сравнении рассчитываются забитые, пропущенные голы и разница?

GF and GA are season totals rebuilt from completed home/away match scores. GD is GF − GA. They come from the optional football-data.co.uk match enrichment, not from the Champagne Index player totals.

How are attacking and defensive ranks calculated?

Как рассчитываются атакующий и защитный рейтинги?

Within one league and season, attack rank orders goals scored per match from highest to lowest; defense rank orders goals conceded per match from lowest to highest. SQL RANK gives tied rates the same rank and leaves a gap after ties. They describe scores, not tactics or chance quality.

Do we have xG or can we estimate it?

Есть ли в данных xG или можем ли мы его оценить?

The source has no provider xG field. The comparison instead shows observed shots and shots on target. A separate estimate multiplies shots on target by the league-season average goals-per-shot-on-target rate; it is an experimental volume proxy with no shot-location or chance-quality input, not xG.

What is the difference between the Champagne Index and attack rank?

Чем атакующий рейтинг отличается от Champagne Index?

The Champagne Index is recorded player goals plus assists divided by a Transfermarkt matches proxy for a club-season. Attack rank uses actual completed league scorelines, sums team goals and ranks goals per match within the league-season. Their sources, coverage and units differ.

How did you check the goalkeeper defensive totals?

Как вы проверили вратарские показатели по пропущенным?

We compared Transfermarkt goalkeeper goals-conceded totals with the independent match-result totals on 388 adequately covered top-five club-seasons. They match exactly in 366 (94.3%). Clean sheets match only 269 (69.3%), so the scoreline source remains the basis of league ranks.

What does the defensive-spend analysis show?

Что показывает анализ расходов на защитников?

The Pearson correlation between defender transfer fees and next-season change in the goalkeeper-source conceded-goal rate is r=0.004047, p=0.426997, n=38,531. The sample does not support a useful linear association. It is descriptive, not causal. A separate match-level check finds teams with a red card concede 0.341 more goals on average in the same match, but match context confounds that association.

Why keep the player-source defensive values separate from scoreboard goals?

Почему defensive-показатели по игрокам отделены от итогов матчей?

The sources use different records and denominators. Goalkeeper conceded totals agree with scorelines in 94.3% of adequately covered seasons, while clean sheets agree in 69.3%. The UI therefore uses match scores for GF, GA, GD and ranks; player-source totals remain a quality cross-check. Shots and corners are also observed match statistics, not chance-quality measures.

Is the shot-based estimate the same as xG?

Оценка по ударам — это xG?

No. It multiplies a team’s shots on target by the league-season average goals per shot on target. It has no shot-location, distance, body-part, pressure or goalkeeper-position input. It is a rough volume benchmark and must be called an SOT-based proxy, not provider xG.

What does the Poisson attack/defense rank tell us?

Что показывают Poisson-ранги атаки и обороны?

The model fits attack, defense and a shared home-advantage term to the selected league-season scores. Larger fitted attack and defense parameters mean stronger estimated attack and defense. Its ranks are descriptive and in-sample; the separate chronological RandomForest evaluation is the project’s held-out prediction check.

Did the added defensive features improve the model?

Улучшили ли защитные признаки модель?

On the same chronological split, R^2 moves from 0.1271 with the five-feature baseline to 0.1502 after adding goalkeeper conceded rate, discipline and net spend; MAE moves from 0.4856 to 0.4788 on 5,585 held-out rows. The gain is modest, and it does not isolate a causal effect of defense.

Do red cards cause more goals to be conceded?

Красные карточки приводят к большему числу пропущенных?

The local sample shows a same-match difference of +0.341 conceded goals per team-match (1,619 sent-off vs 16,195 others; Welch p < 0.0001). This is observational: stronger opponents, match state and timing can affect both red cards and goals. It is not a causal estimate.

Какие фразы не использовать

- Очные ассисты или xG доступны в матчевом источнике.
- Модель обучается заново при каждом движении слайдера.
- Успешный HTTP 200 одной страницы доказывает корректность всех данных.

> Роль — учебная ответственность. Не заявляй личное авторство или самостоятельную проверку, если этого не было. Покажи ровно те изменения и команды, которые выполнял(а) лично.